Wireless Access Electromagnetics Lab 2

Sheharyar Ahmed Nizamani
S329580

23-04-2026

Design Specifications and Substrate Properties

- The objective was to design a microstrip T-junction power divider centered at f = 5.8GHz with an output power ratio P3/P2 = 1.25. The design was implemented on an FR4 substrate with the following parameters:

Er: 4.45

h: 1.55 mm

t: 0.035 mm

The 3 reference impedances in the homework 2 and their geometries are listed below:

Z_{inf1} = 50	l = 10	W = 2.75
Z_{inf2} =75	l = 6.92	W = 1.37
Z_{inf3} = 67.08	l = 6.92	W = 1.73

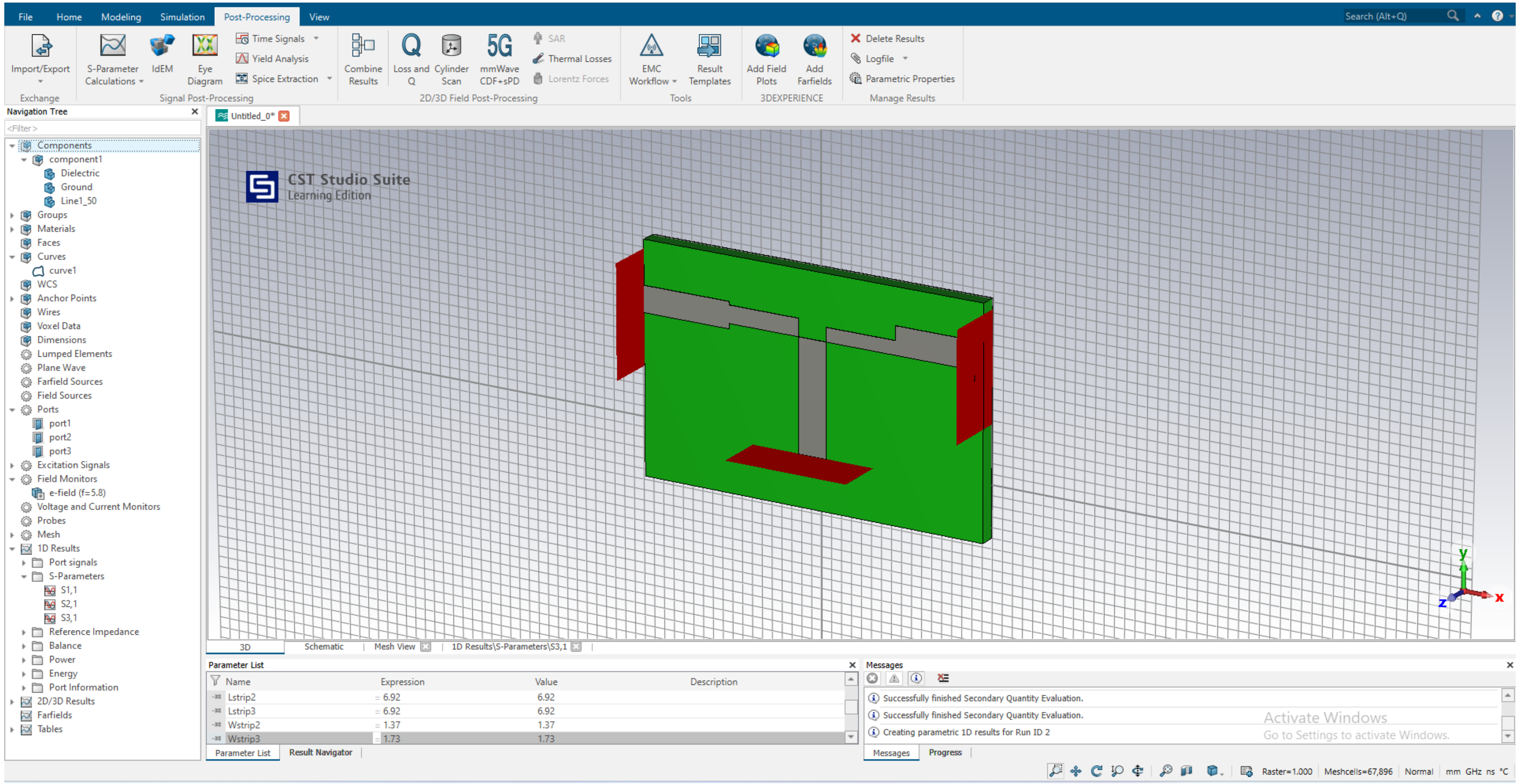

Fig 1: The CST geometry

This 3D model shows the microstrip T-Junction. It includes 3 waveguided ports, and 2 output branches with asymmetric widths. After setting up the solver, I obtained the following results:

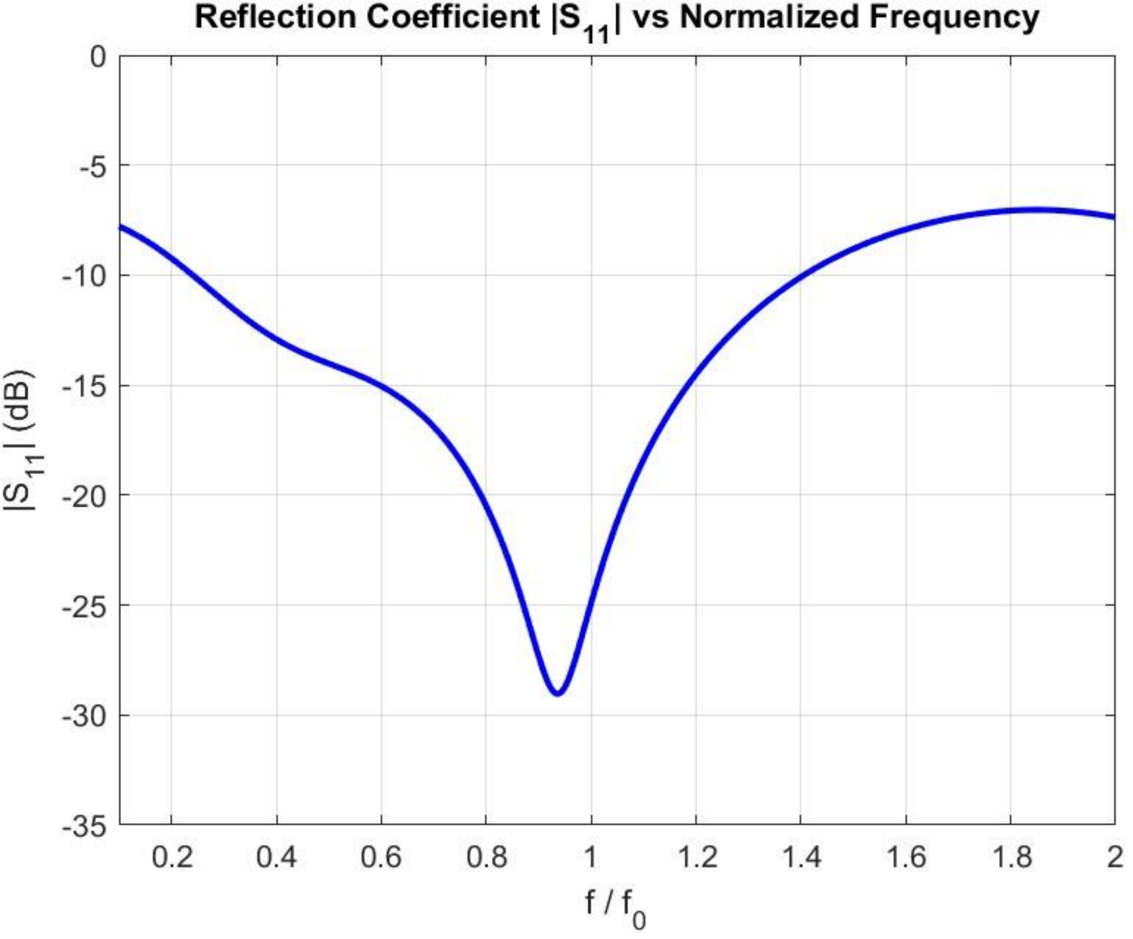

Fig 2: S_{11} v Normalized Frequency

The simulated S_{11} shows a significant resonance at the 5.8GHz design frequency, and the loss at the design frequency is approximately –25dB, which means the junction is perfectly matched with very low reflected power.

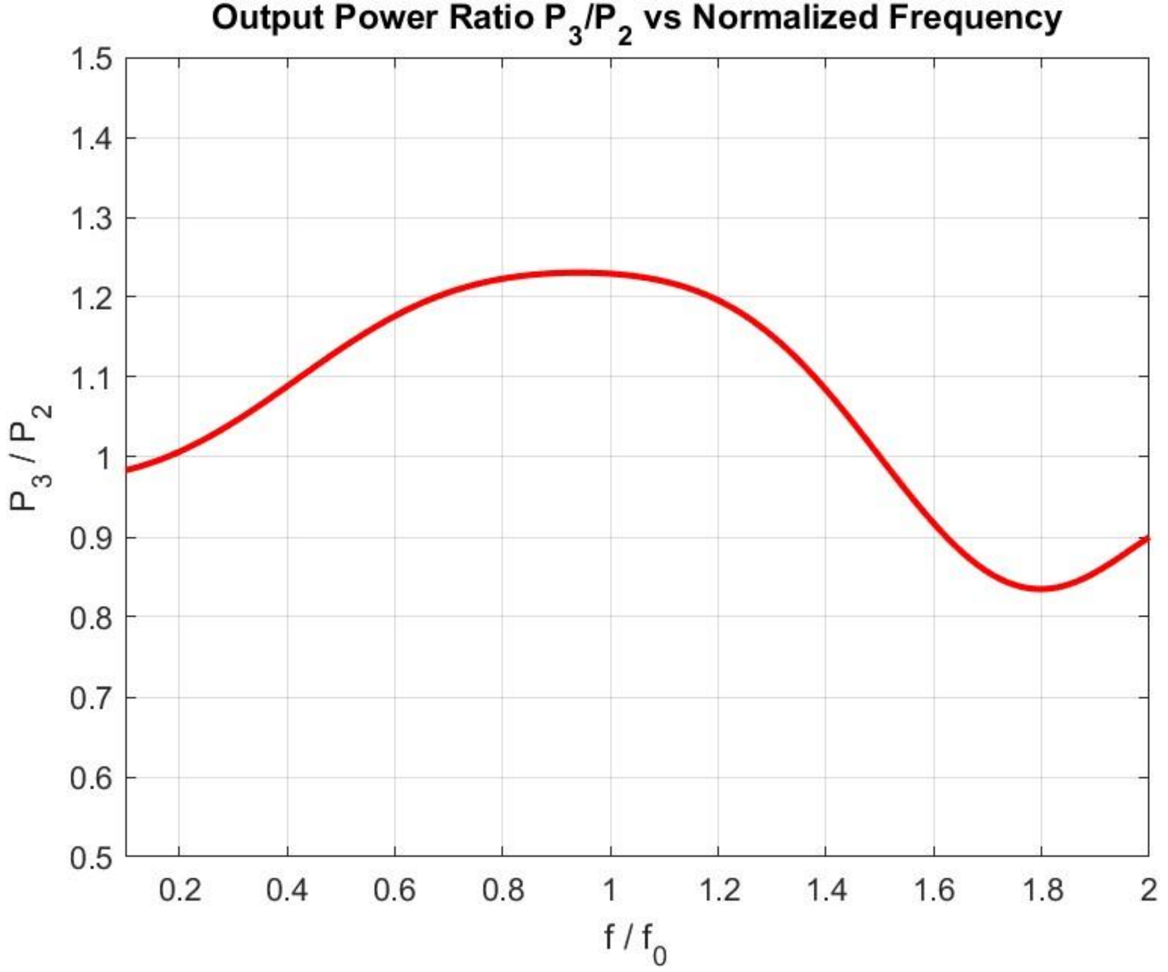

Fig 3: P_3/P_2 v Normalized Frequency

At the center frequency f/f0 = 1, the power ratio is approximately 1.27. This is very close to the design target ratio of 1.25, this shows that my calculated transformer widths are in line with the requirement.

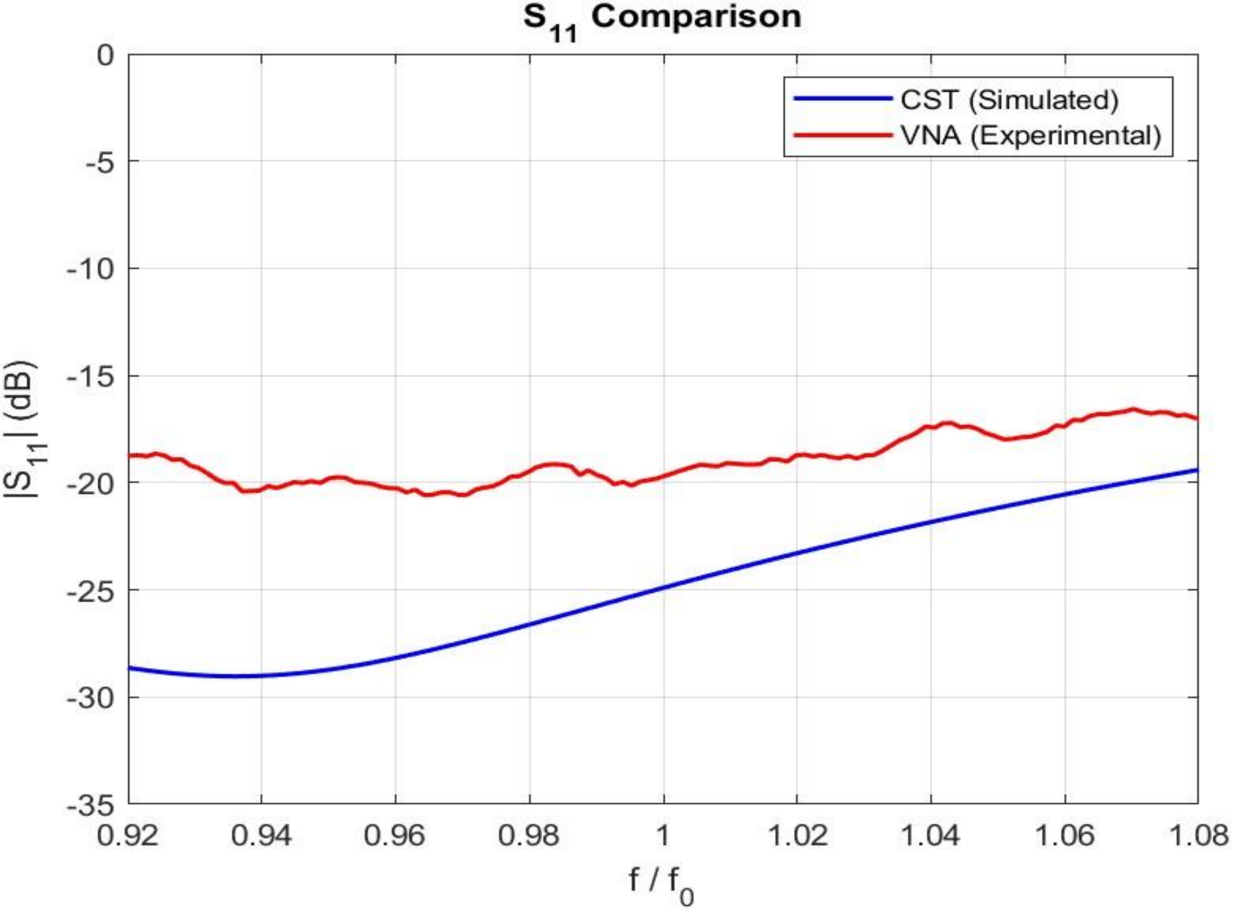

Fig 4: CST & VNA comparison of S_{11}

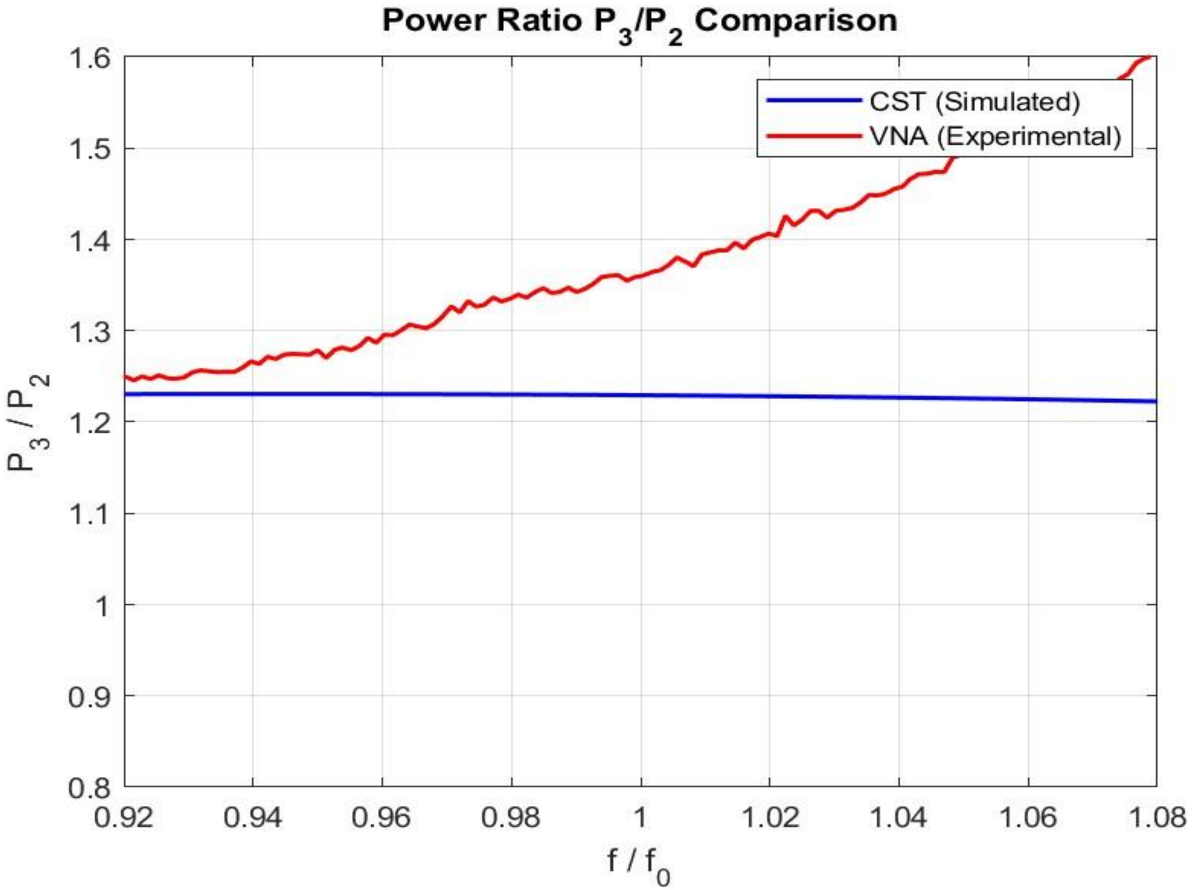

Fig 4: CST & VNA comparison of power ratio

The VNA and CST curves follow the same trend. The VNA dip isn't as deep as the simulation, this is because the real device has SMA connectors, solder, and cable losses that aren't accounted for in the ideal CST model.

Both curves stay close to the target ratio near 5.8 GHz. The VNA ratio is slightly higher than the CST one (about 1.35 vs 1.27). This could be due to small asymmetries from manual soldering or the manufacturing tolerances of the FR4 substrate.